MILITARY SPECIFICATION

BOMB, FRAGMENTATION, DUMMY BLU-61A(D-1)/B

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Bomb designated as Bomb, Fragmentation, Dummy BLU-61A(D-1)/B.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions

FSC 1325

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MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F-9256740 - Bomb, Fragmentation BLU-61A/B

PUBLICATIONS

ARMY

9256740 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite, "49 CFR 171-17, (latest revisions)").

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on drawing 9256740, all associated drawings and with all requirements specified in applicable specifications.

3.3 Loading

3.3.1 Just before loading, the base and top section assemblies shall be examined for foreign matter. No solid filler shall be present in the base or top sections before loading.

3.3.2 No solid filler shall be added during pouring or cooling. The interlock surfaces shall be free of caked filler. A chemical cleaner shall not be used.

NOTE: Clean scrap produced in regular operations may be remelted and used, provided it complies with the grade of filler specified except for form and viscosity. Clean scrap is defined as scrap explosive formed in kettles, pails, risers, formers, and loading machines used in pouring, casting and drilling operations.

3.4 Surface condition of charge.-Small chips and voids are allowed on the surface of the filler provided their depths do not exceed one-eighth (1/8") inch, and their combined area does not exceed one (1) square inch. Chipped areas around breakouts shall be included as a portion of the allowed total void area.

3.5 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract, the specification, or as requested by the Contracting Officer. (see 6.4).

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contamination by the cleaning agent. All required markings shall be neat and sharply defined.

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3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of compliance on all material purchased by the contractor when such material is controlled by Government or other required specification referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2 (see 6.1). The first article sample shall consist of the following:

a. Eighty (80) springs
b. One hundred twenty-six (126) Shell, Lower Loaded Inert (9256736)
c. One hundred twenty-six (126) Shell, Upper Loaded Inert (9256737)
d. Two hundred (200) Bomb, Fragmentation BLU-61A(D-1)/B (9256740)

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing processes, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed.

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4.2.2.1 Quantities for initial inspection.-After classification of defects and other inspections have been performed as specified in 4.3.2, the following quantities shall be subjected to the indicated tests.

4.2.2.2 Specific gravity of charge.-Twenty six (26) loaded assemblies shall be subjected to this test.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturen, in one unchanged process using the same materials and methods, in accordance with the same drawing, same drawing revisions, same specifications, and same specification revisions. Changes to the process, specification, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

Unless otherwise approved by the Contracting Officer, the inspectio lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots components or subassemblies, other than

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the items of delivery, shall be homogeneous and of a size mutally convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that, inspection for Critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government, and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect except where 100% inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Spring (See Dwg. 8833582)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.50 Percent		
201.	Thickness	Gage	01001
202.	Height after forming	Gage	01002
203.	Evidence of poor workmanship (see 3.6)	Visual	01003

4.3.2.2 Shell, Upper Loaded Inert (9256737)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00%		
101.	Depth, shell to bottom of smaller cavity	Gage	02001
102.	Depth, top to bottom of smaller cavity	Gage	02002
103.	Concentricity of larger cavity with shell outer diameter	Gage	02003
104.	Depth to charge within 2.65 diameter	Gage	02004
105.	Charge above flush outside 2.65 diameter	Gage	02005
106.	Smaller cavity diameter, min.	Gage	02006
107.	Larger cavity diameter, min.	Gage	02007
Minor			
201.	Larger cavity diameter, max.	Gage	02008
202.	Marking missing	Visual	02009

4.3.2.3 Shell, Lower Loaded Inert (9256736)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 1.00 Percent		
101.	Depth, shell diameter to bottom of second largest cavity	Gage	03001
102.	Height of second largest cavity	Gage	03002
103.	Depth to charge within 2.65 dia.	Gage	03003
104.	Concentricity of largest cavity with respect to shell diameter	Gage	03004
105.	Concentricity of second largest cavity with respect to largest shell diameter	Gage	03005
106.	Diameter, largest cavity, min	Gage	03006
107.	Diameter, second largest cavity, min.	Gage	03007
108.	Charge above flush outside 2.65 diameter	Gage	03008
Minor:	AQL 1.50 Percent		
201.	Diameter, large, cavity, max	Gage	03009
202.	Diameter, second largest cavity, max.	Gage	03010
203.	Marking missing	Visual	03011

4.3.2.4 Bomb, Frag, BLU-61A(D-1)/B prior to joining top and base loading assemblies (see dwg. 9256740)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Dummy Fuze missing	Visual	04001
102.	Spring missing or presence of more than one spring	Visual	04002
103.	Incorrect application of adhesive or adhesive missing	Visual	04003
104.	Felt Pad missing	Visual	04004

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Minor:	AQL 1.00 Percent		
201.	Sealing compound missing or excessive	Visual	04004
202.	Foreign matter	Visual	04005
203.	Orientation of spring w/respect to filler	Visual	04006

4.3.2.5 Bomb, Fragmentation BLU-61A(D-1)/B (See Dwg. 9256740)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Gap at crimped strap excessive	Gage	05001
102.	Strap torn, broken, cracked or improperly crimped	Visual	05002
103.	Bomb cracked or deformed (especially at flange)	Visual	05003
104.	Flutes (ribs) not alighed	Visual	05004
105.	Diameter across bomb at strap	Gage	05005
106.	Marking illegible or missing	Visual	05006
107.	Distance across crimp	Gage	05007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	05008

4.3.2.6 Upper and Lower shell assemblies.-Prior to loading, the upper and lower assemblies shall be examined for conformance to the requirements of 3.3.1. This examination shall be applied, using a CSP-1 continuous sampling plan (MIL-STD-1235) at an AQL of 1.0% or a plan assuring equivalent protection. Code No. 06001. After pouring, the assemblies shall be examined for conformance to 3.3.2 using the same sampling plan. Code No. 07001

4.3.2.7 Surface condition.-After loading, but prior to assembly of the Fragmentation Bomb, upper and lower shall assemblies shall be examined for conformance to the requirements of 3.4 using, a CSP-1 continuous sampling plan (MIL-STD-1235) at an AQL of 1.00%, or a plan assuring equivalent protection. Code No. 08001

4.3.3 Testing

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4.3.3.1 Specific gravity of charge.-A sample of three (3) loaded assemblies shall be selected from each four (4) hours production and tested as specified in 4.4.2. If any sample fails to conform to the drawing requirements, the four (4) hour's production shall be rejected. Code No. 10001.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9241951 and Equipment Lists (EL's) referenced theron identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacturer, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

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4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall, when required by the Government, make available all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 Specific gravity of charge.-The representative sample metal part hemisphere shall be weighed empty, then filled with mercury and re-weighed. The mercury shall then be removed and each hemisphere loaded with filler under conditions identical in all respects with those employed in loading regular hemispheres. The fuze cavity, or fuze and booster assembly cavity shall be drilled or formed and the loaded sample hemisphere re-weighed. The drilled cavities shall be filler with mercury and the hemisphere re-weighed. The specific gravity shall be calculated as follows:

$$\text{Specific gravity} = \frac{13.55}{(W_m - W_e) - (W_f - W+)} \quad (W+ - We)$$

Where:

We = weight of metal part
W_m = weight of part filled with mercury
W+ = weight of loaded and drilled assembly
W_f = weight of loaded, mercury and filled cavity section

4.4.2 Alternate method for specific gravity of charge inspection.-The representative sample metal part hemisphere shall be weighed in air (without filler) and then weighed in water (without explosive). The hemisphere is then loaded with filler under conditions identical in all respects with those employed in loading regular hemispheres. The hemisphere shall be drilled and then weighed in air, and in water. The specific gravity will be calculated as follows:

$$SG = \frac{WL - We}{(WL - We) - (WWL - WWe)}$$

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Where:

We = weight of empty hemisphere part in air
WL = weight of loaded and drilled assembly in air
WWe = weight of empty hemisphere part, in water
WWL = weight of loaded and drilled assembly, in water

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing.-Not applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.6.

6.3.1 Equipment Lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock Numbers of existing equipment design, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4, for all equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6. Action required by the contractor with respect to the use of commercial equipment is detailed in 6.3.5.

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6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army design.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classification; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army design, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections (See 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.4 Contractor designs.-Contractor designs are designs of inspections equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit a copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.6.

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6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors, normally, are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of the fixturing shall be considered as part of the total design, and 6.3.4 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.-All submitted designs shall reference the applicable five-digit code number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3. All other designs or inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian
Army-MU

Preparing Activity:
Army-MU

Project Number: 1325-A416

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